Replication with Language Models

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Why Do So Many Studies Fail to Replicate?



The New York Times

Opinion

GRAY MATTER

Why Do So Many Studies Fail to Replicate?

By Jay Van Bavel

May 27, 2016

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Van Bavel, "Why Do So Many Studies Fail to Replicate?" The New York Times, Gray Matter, 27 May 2016.

In 2015, the Open Science Collaboration tried to replicate 100 published psychology studies. **About one in three reproduced the original result** (Open Science Collaboration 2015).

A New Kind of Coder, a New Kind of Fragility

The classic replication crisis was about p-hacking, small samples, and publication bias. LLM-based research has a different, newer problem.

> *LLMs "at first appear to enable replication as the discipline commonly understands it — yet in practice they frequently fall short." (Barrie, Palmer, and Spirling 2024)*

Unlike a fixed dataset or a pinned software version, a **proprietary model itself can simply cease to exist** — deprecated, updated, or retired out from under you, with no guarantee the old version is archived anywhere.

The Promise

> Large language models can **annotate text at scale** — coding thousands of manifestos, tweets, or survey responses in minutes, at a tiny fraction of the cost of human coders or crowdworkers.

- Political science has produced **almost no work** on replication and research-transparency standards for LLM-based studies (Barrie, Palmer, and Spirling 2024)
- Meanwhile LLMs are already being used *as if* they were just another (better, cheaper) coder
- This talk asks: what does it mean for an LLM-based study to **replicate** — and can we test that, rather than just hope for it?

Paper 1

Prompt Stability Scoring for Text Annotation with Large Language Models

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Abstract

Researchers are increasingly using language models (LMs) for text annotation, typically relying on prompts that instruct the model to return a given output. LM outputs can nonetheless be vulnerable to minor prompt variations, raising concerns about the reliability and reproducibility of classification routines. Stability is a necessary but not sufficient condition for both accuracy and reproducibility. To diagnose this, we propose the Prompt Stability Score (PSS), an automated procedure for measuring agreement across repeated runs and semantically similar prompt variants, and provide an accompanying Python package, `promptstability`. Using six datasets and twelve outcomes, we generate ∼3.1m annotations (∼300m input tokens) to characterize when prompt-based pipelines are stable versus fragile. We conclude with practical recommendations for applied research.

Title page and abstract, Paper 1 (Barrie, Palaiologou, and Törnberg 2024)

Forthcoming in Political Science Research and Methods.

Paper 2

Replication for Large Language Models

Problems, Principles, and Best Practices for Political Science*

Christopher Barrie† Alexis Palmer‡ Arthur Spirling§

June 5, 2026

Word count: 9736

Abstract

Large Language Models (LLMs) are exciting tools: they require minimal researcher input but make it possible to annotate and generate large quantities of data. Yet there has been almost no systematic research into the reproducibility of research using LLMs. This is a potential problem for scientific integrity. We give a theoretical framework for replication in the discipline and show that LLM work is perhaps uniquely problematic. We demonstrate the problem empirically using a rolling iterated replication design in which we compare crowdsourcing and LLMs on multiple repeated tasks, over many months. We find that LLMs can be accurate, but the observed variance in performance is often unacceptably high. Strict "temperature" control does not resolve these issues. This affects downstream results. In many cases the LLM findings cannot be re-run, let alone replicated. We conclude with recommendations for best practice, including the use of locally versioned 'open' LLMs.

Title page and abstract, Paper 2 (Barrie, Palmer, and Spirling 2024)

Forthcoming in the American Journal of Political Science.

Two Papers, One Question

Two companion papers, both asking whether LLM-based annotation can be trusted to replicate:

- **Paper 1** (Barrie, Palaiologou & Törnberg): a diagnostic **tool** — can you tell, cheaply, whether a pipeline is fragile? (Barrie, Palaiologou, and Törnberg 2024)
- **Paper 2** (Barrie, Palmer & Spirling): an empirical **audit** — repeated LLM replications over a year, tracking what actually breaks (Barrie, Palmer, and Spirling 2024)
- **Shared logic**: both repurpose human inter-coder reliability approaches for a new kind of "rater" — the LLM
- **This talk**: where LLMs sit in replication's landscape, the diagnostic's mechanics, the empirical evidence, a synthesis — then how to run the diagnostic yourself

LLMs Don’t Fit Any Existing Replication Category

Paper 2’s Table 1 places replication practices on two axes — and shows where proprietary LLMs land, alone, in the worst quadrant (Barrie, Palmer, and Spirling 2024):

Robust / system-independent?	Exact replication possible? — No	Exact replication possible? — Yes
No	**(Proprietary) Large Language Models**	**Deterministic Replication** static code, static data *e.g. King (1995)*
Yes	**Stochastic Replication** crowdsourcing, undergrad RAs *e.g. Benoit et al. (2016)*	**Simple, Rule-Based** expert agreed standard *e.g. Bateman et al. (2015)*

“The central problem with replication for LLMs is that the process exhibits the weaknesses of deterministic, stochastic and rule-based replication, without the strengths of any of them.” (Barrie, Palmer, and Spirling 2024)

How Prompt Stability Scoring Works

> Paper 1's trick: treat repeated LLM runs — or reworded prompts — **as if they were multiple human raters** (Barrie, Palaiologou, and Törnberg 2024).

- **Krippendorff's alpha**: $\alpha = 1 - D_o/D_e$ — 1 is perfect agreement, 0 is chance-level
- **Intra-PSS**: identical prompt run **30 times**; α computed across repeated runs
- **Inter-PSS**: **10 paraphrased variants** per "temperature" (0.1–5.0); α across variants
- **Confidence intervals**: nonparametric bootstrap — resample records, recompute α, repeat

Finding 1: One Voter, Three Nearly Identical Prompts

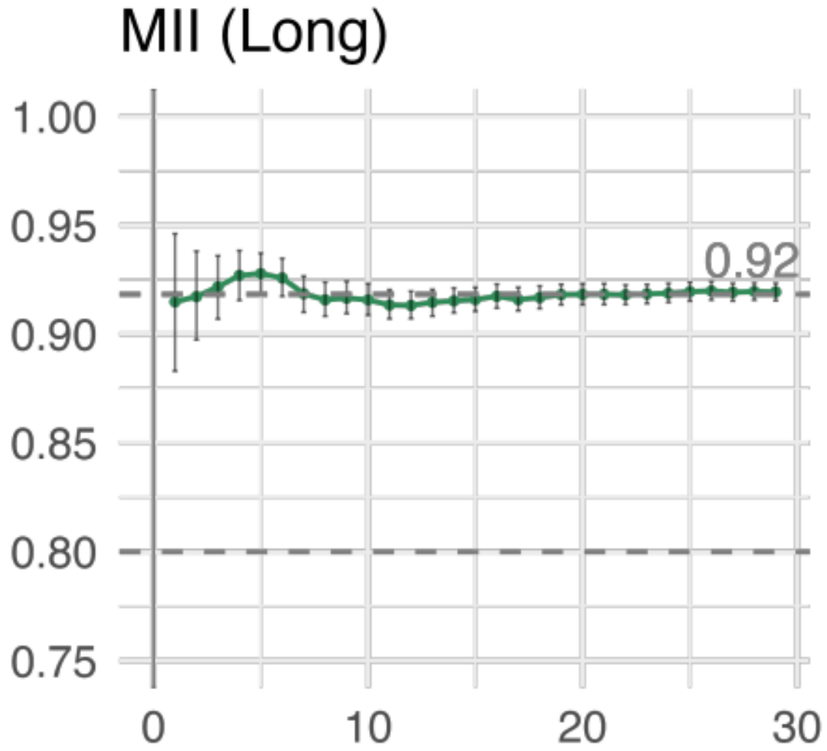
The task: guess how one real voter voted, from minimal biographical information (Barrie, Palaiologou, and Törnberg 2024).

> *Racially, I am Native American. I like to discuss politics with my family and friends. I do not attend church. I am 59 years old. I am a man. I am very interested in politics.*

1. **Original prompt** — "…I want you to tell me how you think they voted. [Respond 0 for Biden, or 1 for Trump…]" → **Output: 1**
2. **Reworded, trivially** — "I want you to tell me how you voted in the election. [Respond 0 for Biden, or 1 for Trump…]" → **Output: 0**
3. **Reworded, trivially, again** — "I want you to tell me how you voted. [Respond 0 for Biden, or 1 for Trump…]" → **Output: 1**

Finding 1, at Scale: Prompt Wording Changes the Labels

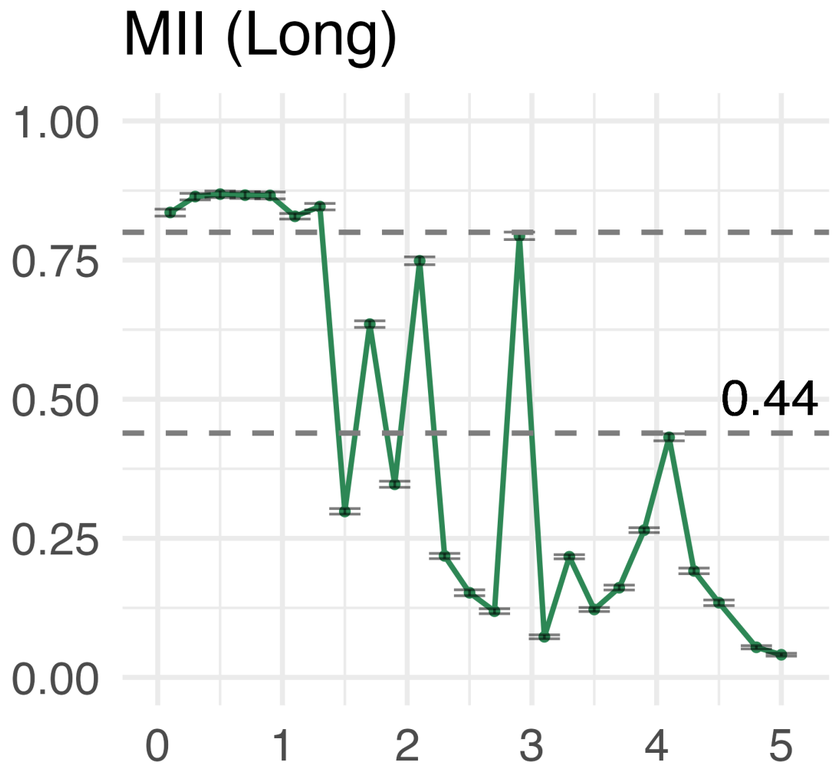
Same real outcome (“MII, Long” — UK survey most-important-issue coding), scored with the statistic just described. Same prompt, repeated 30 times:



Paper 1, Figure 1 (intra-PSS), MII (Long) (Barrie, Palaiologou, and Törnberg 2024)

Finding 1 (continued): Reworded, Across “Temperature”

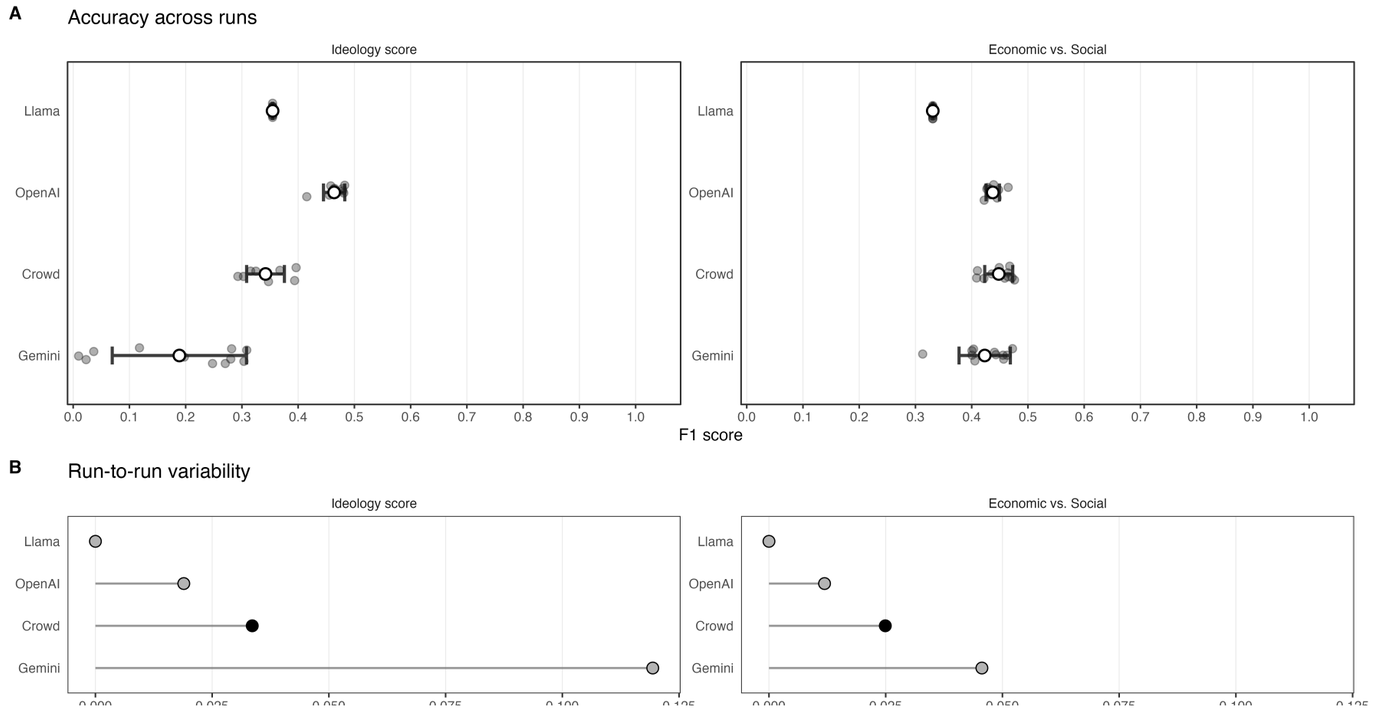
Same outcome (MII, Long), same statistic — now the prompt is paraphrased at increasing “temperature” (0–5) instead of repeated verbatim:



Paper 1, Figure 2 (inter-PSS), MII (Long) (Barrie, Palaiologou, and Törnberg 2024)

Finding 2: Model & Version Choice Changes Accuracy *and* Variance

Wording is one axis of fragility. Paper 2 tests another: crowdworkers, GPT, Gemini, and Llama on the same manifesto task, across eleven monthly iterations (Barrie, Palmer, and Spirling 2024).

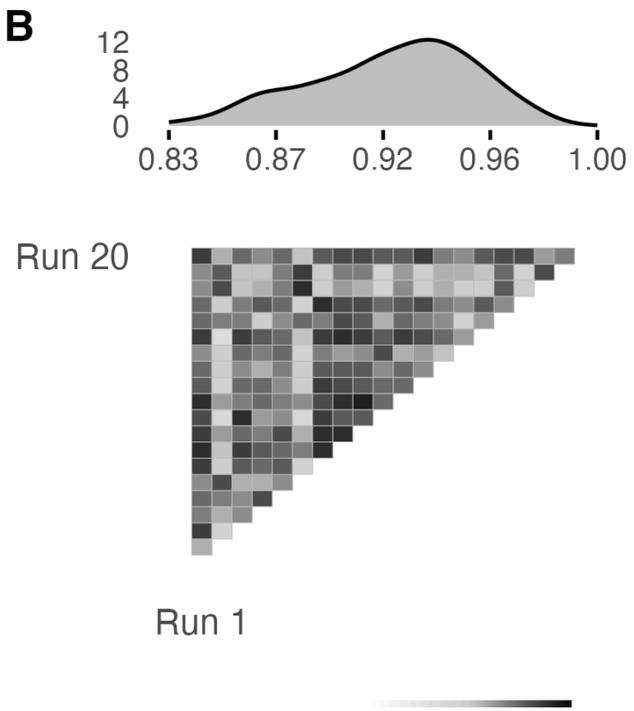


Paper 2, Figure 1, Panels A–B ("Manifestos, static") (Barrie, Palmer, and Spirling 2024)

Not hypothetical: in February 2026, OpenAI deprecated and removed a popular model, `o1-mini` *— mid-study, for someone relying on it (Barrie, Palmer, and Spirling 2024).*

Finding 3: “Temperature = 0” Is Not Determinism

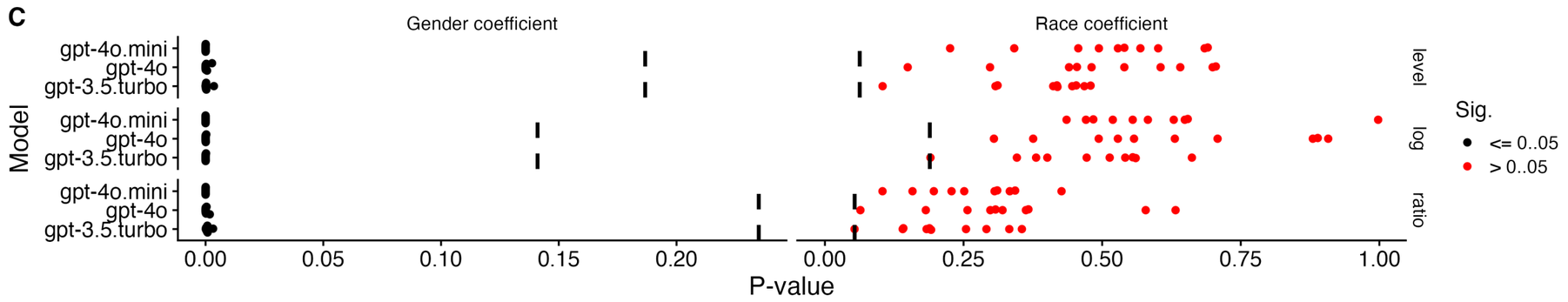
The intuitive fix for instability is “set temperature to 0.” Paper 2 tested this directly (Barrie, Palmer, and Spirling 2024).



Paper 2, Figure 2, Panel B (`gpt-4o-2024-08-06`, temp = 0) (Barrie, Palmer, and Spirling 2024)

Finding 4: Annotation Choices Flip Downstream Conclusions

So far, every finding has been about label reliability. Does any of it change the conclusions a paper actually draws? Paper 2 re-coded Hopkins et al.'s ~6.6k identity-cued headlines with three OpenAI models, then re-ran the original regressions (Barrie, Palmer, and Spirling 2024).



Paper 2, Figure 5, Panel C (P-values, original vs. re-estimated) (Barrie, Palmer, and Spirling 2024)

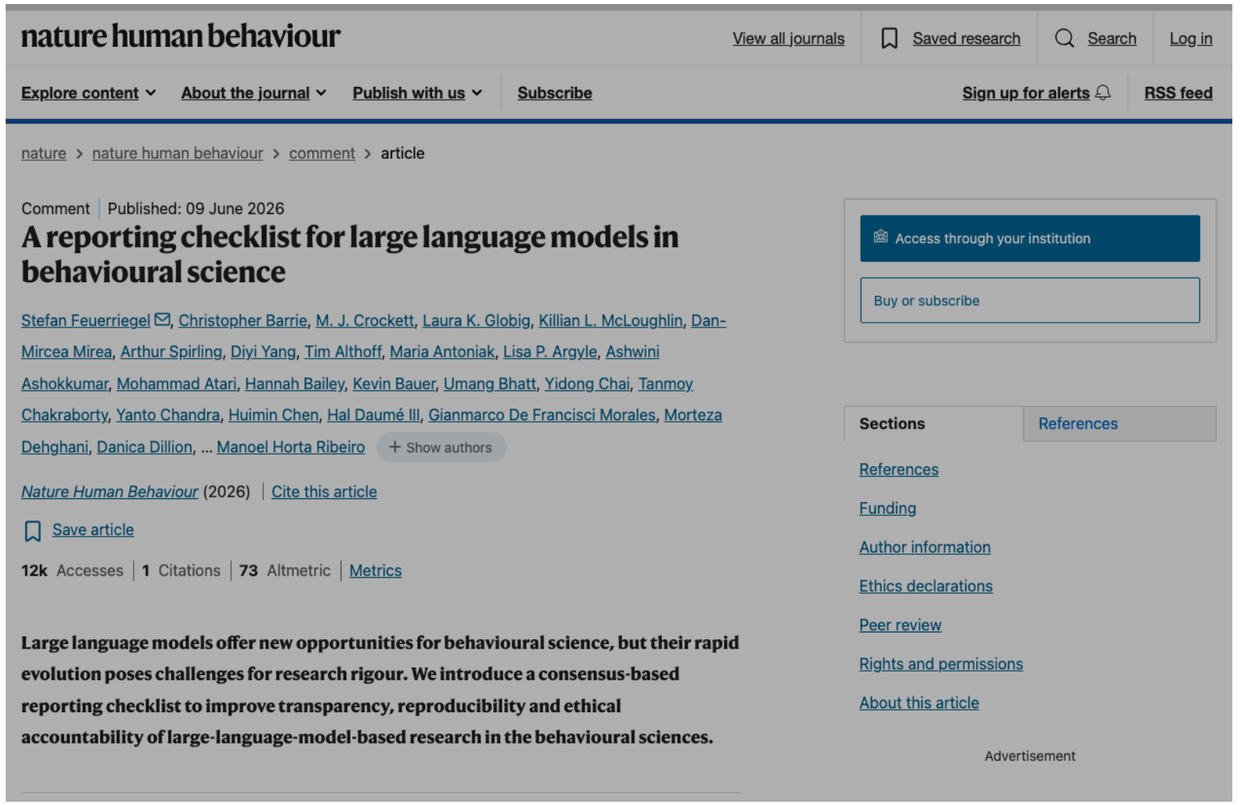
Gender: not significant originally, **significant for every model** tested. Race: the reverse.

Putting It Together

A practical robustness workflow combining both papers' tools:

1. **PSS pre-check** — run intra-/inter-PSS before investing further
2. **Validate a subsample** against expert or gold-standard codes
3. **Sweep models and versions** — does the finding survive?
4. **Prefer open, versioned models** where the task allows it
5. **Document everything** — model, version, prompt, settings, date run

A Field-Wide Checklist



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A reporting checklist for large language models in behavioural science

Stefan Feuerriegel, Christopher Barrie, M. J. Crockett, Laura K. Globig, Killian L. McLoughlin, Dan-Mircea Mirea, Arthur Spirling, Diyi Yang, Tim Althoff, Maria Antoniak, Lisa P. Argyle, Ashwini Ashokkumar, Mohammad Atari, Hannah Bailey, Kevin Bauer, Umang Bhatt, Yidong Chai, Tanmoy Chakraborty, Yanto Chandra, Huimin Chen, Hal Daumé III, Gianmarco De Francisci Morales, Morteza Dehghani, Danica Dillion, ... Manoel Horta Ribeiro + Show authors

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Large language models offer new opportunities for behavioural science, but their rapid evolution poses challenges for research rigour. We introduce a consensus-based reporting checklist to improve transparency, reproducibility and ethical accountability of large-language-model-based research in the behavioural sciences.

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Feuerriegel, Barrie, Crockett, et al. (2026), "A reporting checklist for large language models in behavioural science," Nature Human Behaviour (Feuerriegel et al. 2026). Click to open the article.

Try It Yourself: the `promptstability` Package

> Both PSS statistics in this talk — intra-PSS, inter-PSS, and the bootstrap confidence intervals around them — are one `pip install` away, and work with any annotation function you can write a Python wrapper for.

- `pip install promptstability` — tested against OpenAI and local models (e.g. via Ollama)
- Source & full API: `github.com/cjbarrie/promptstability`
- Docs: `promptstability.readthedocs.io`
- The appendix walks through its core functionality, with a real example run

Appendix: Inside the `promptstability` Package

> The package that computes every PSS number in this talk. This appendix walks through installing it, its core API, and a real example run.

Appendix: Running Intra- and Inter-PSS

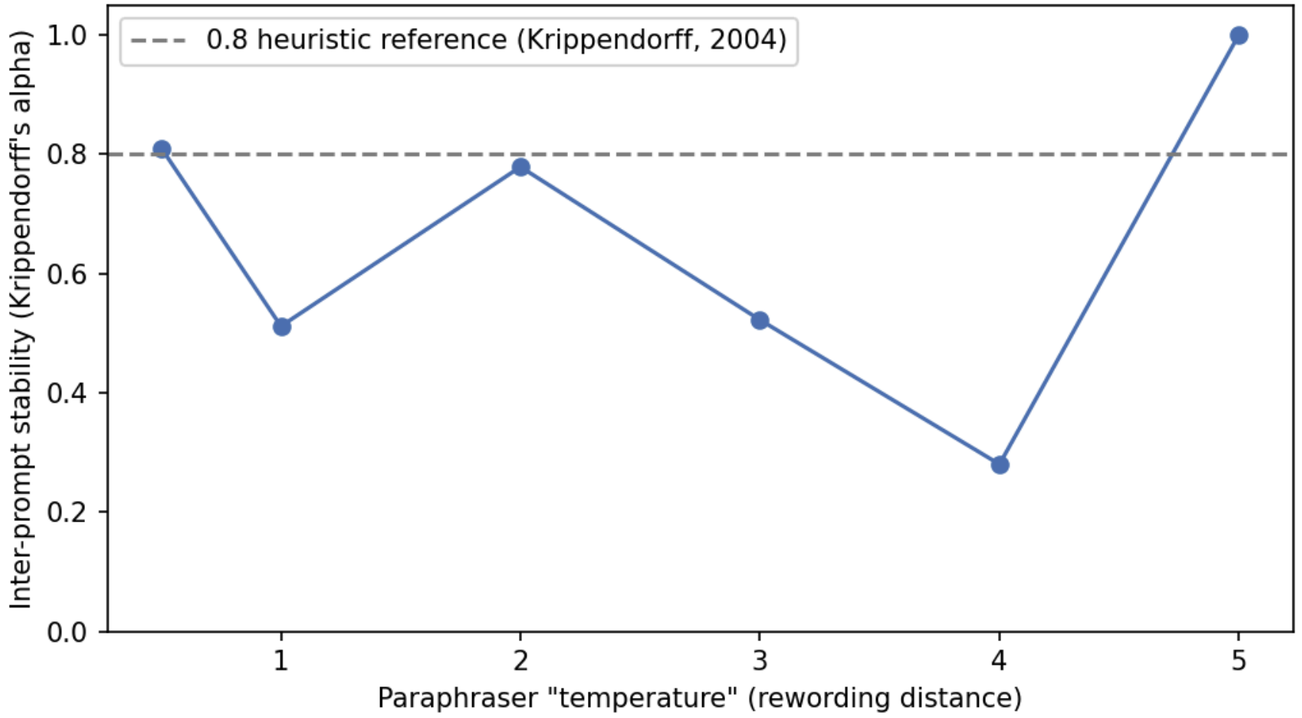
A minimal example — any function that takes `(text, prompt)` and returns a label works as the annotation function, whether it calls OpenAI or a local Ollama model:

```python
from promptstability import PromptStabilityAnalysis

def annotate_ollama(text, prompt):
    r = ollama.chat(model="llama3.2:3b",
        messages=[{"role": "system", "content": prompt}, {"role": "user", "content": text}],
        format={"type": "object", "properties": {"label": {"type": "integer", "enum": [0, 1]}}},
        options={"temperature": 0.1})
    return r["message"]["content"]

psa = PromptStabilityAnalysis(annotation_function=annotate_ollama, data=data,
                              parse_function=lambda r: json.loads(r)["label"])
ka_intra, _ = psa.intra_pss(ORIGINAL_PROMPT, PROMPT_POSTFIX, iterations=5)
ka_inter, _ = psa.manual_inter_pss("prompt_variants.csv")
```

Appendix: Real Output From the Package



Appendix: API Surface

- `PromptStabilityAnalysis(annotation_function, data, parse_function=)` — wraps any annotation function (OpenAI, Ollama, local models) as a PSS-ready analysis object
- `.intra_pss(original_text, prompt_postfix, iterations=)` — identical prompt, run n times; α across runs
- `.inter_pss(original_text, prompt_postfix, nr_variations=, temperatures=)` — built-in PEGASUS paraphraser generates variants automatically across a temperature sweep
- `.manual_inter_pss(csv_path)` — same statistic, but scored against your own hand-written prompt variants
- Bootstrap confidence intervals and nominal- vs. interval-distance metrics (categorical vs. ordinal outcomes) are handled automatically based on the outcome type
- `load_example_data()` and `get_api_key()` helpers for quick starts

References

Barrie, Christopher, E. Palaiologou, and P. Törnberg. 2024. “Prompt Stability Scoring for Text Annotation with Large Language Models.” *Political Science Research and Methods*.

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